

# FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.3 CHEMICAL BONDING: FINAL EXPLANATION

#### Student Assessment Document

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|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use what you observed and learned in Lessons 1 to 8 to answer the driving question: Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently, and what is happening between the atoms inside each one that controls every property we can measure? Write your answers in the blank space under each prompt. Use full sentences and particle-level words such as ions, lattice, delocalised electrons, covalent bonds and Van der Waals forces. For each claim, give the evidence from our tests and the reasoning that links the evidence to the particles. You may use a labelled sketch where a prompt allows it. Refer to your science journal and the Driving Question Board.

#### Part 1: Table Salt (Giant Ionic Structure)

The coastal table salt did not light the bulb as a dry solid, but it did light the bulb when dissolved in water. It also has a very high melting point of over 800 °C and shatters instead of bending. (a) Name the type of bond and the structure in solid NaCl and describe how it forms, using electrons. (b) Using a claim, evidence and reasoning (CER) structure, explain why dry salt does not conduct but salt solution does. (c) Explain why salt needs such a high temperature to melt. (d) Explain why a crystal of salt breaks instead of bending.

*Write your answer here:*

## Part 2: Aluminium (Giant Metallic Structure)

The aluminium wire from the sufuria handle was shiny, bent without breaking, and conducted electricity immediately as a solid. It melted at  $660\text{ }^{\circ}\text{C}$ , which is lower than salt. (a) Describe the bonding and structure inside aluminium. (b) Explain why aluminium conducts when solid, while solid NaCl does not. (c) Explain why aluminium bends instead of shattering. (d) Aluminium has strong bonding, yet it melts at a lower temperature than salt. What does this tell you about comparing the two structures?

*Write your answer here:*

### Part 3: Graphite (Giant Atomic / Layered Covalent Structure)

The graphite rod from the AA battery is a non-metal made only of carbon, yet it conducts electricity as a solid. It is dull grey, snaps instead of bending, and leaves marks on paper. (a) Describe the bonding within a graphite layer and between layers. (b) Explain why graphite conducts even though carbon is a non-metal. (c) Explain why graphite leaves a mark on paper. (d) A classmate says 'only metals conduct electricity'. Use the evidence from our tests to respond.

*Write your answer here:*

#### Part 4: Comparing Structures and Predicting New Substances

(a) Complete this table in your answer by writing the structure type, the particles that are free to move (if any) in the solid, and one reason for each substance's melting point or mechanical behaviour: NaCl, aluminium, graphite. (b) In Lesson 7 you predicted the properties of two new substances: iodine crystals (used in water-purification tablets sold in Nairobi) and iron in a Jua Kali welding rod. Choose ONE and predict its conductivity, melting point and mechanical behaviour, giving a reason from its structure.

*Write your answer here:*

#### Part 5: The Complete Story and Revising Your Thinking

Write a final explanation of at least one paragraph that answers the driving question: Why do the sufuria aluminium, the graphite pencil and the table salt look similar as solids but behave so differently, and what is happening between the atoms that controls their properties? Your answer must (a) include all three substances, (b) link bond type to structure to properties, and (c) say how your thinking changed from your first prediction in Lesson 1. Then give one real-life reason for a material choice in Kenya based on what you learned.

Write your answer here:

| RUBRIC                                         |                                                                                                                                                                                        |                                                                                                                 |                                                                                                               |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Criterion                                      | Excellent (4)                                                                                                                                                                          | Proficient (3)                                                                                                  | Developing (1–2)                                                                                              |
| Linking bond type and structure to properties  | Correctly identifies ionic, metallic and layered covalent bonding for all three substances and clearly links each structure to conductivity, melting point and mechanical behaviour.   | Identifies the bond type for all three substances and links most properties to structure, with one or two gaps. | Names some bond types but describes properties without linking them to structure, or mixes up the substances. |
| Particle-level reasoning about conductivity    | Explains conduction in terms of freely moving ions or delocalised electrons and states why they are free or fixed in each substance, including graphite and dry versus dissolved salt. | Explains conduction using mobile ions or electrons for most substances, but with minor omissions.               | Relies on rules such as 'metals conduct' or 'ionic does not conduct' without describing particles.            |
| Use of evidence (CER)                          | Supports every claim with specific evidence from our tests (for example the bulb lighting, 660 °C, over 800 °C, brittle versus bends) and clear reasoning.                             | Gives claims with some evidence from the tests, with reasoning that is sometimes incomplete.                    | Gives claims with little or no evidence, or evidence without reasoning.                                       |
| Applying ideas to new substances and real life | Accurately predicts the properties of iodine or iron from its structure and gives a sound Kenyan real-life reason for a material choice.                                               | Makes a mostly correct prediction with partial justification and a simple real-life link.                       | Prediction is incorrect or has no reasoning, and the real-life link is missing or unclear.                    |

|                                      |                                                                                                                                                  |                                                                                 |                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Scientific vocabulary and reflection | Uses terms such as lattice, electrostatic, delocalised and Van der Waals precisely, and reflects clearly on how thinking changed since Lesson 1. | Uses most terms correctly and gives a brief reflection on how thinking changed. | Uses few scientific terms correctly and gives little or no reflection. |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------|

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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